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ELECTRONIC DEVICE WHICH TRANSMITS MEDIA CONTENT, AND OPERATING METHOD THEREOF

Abstract

An electronic device is provided. The electronic device includes a communication circuit, memory, including one or more storage media, storing instructions, and at least one processor communicatively coupled to the communication circuit and the memory, wherein the instructions, when executed by the at least one processor, individually or collectively, cause the electronic device to transmit, via the communication circuit capability information of the electronic device and a request for transmitting capability information of an external electronic device to the external electronic device, obtain, via the communication circuit, the capability information of the external electronic device from the external electronic device, based on the obtained capability information of the external electronic device, generate preview content to be output from the external electronic device, and transmit, via the communication circuit, the generated preview content to the external electronic device.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is a continuation application, claiming priority under 35 U.S.C. § 365(c), of an International application No. PCT/KR2023/019067, filed on Nov. 24, 2023, which is based on and claims the benefit of a Korean patent application number 10-2022-0160328, filed on Nov. 25, 2022, in the Korean Intellectual Property Office, and of a Korean patent application number 10-2022-0171375, filed on Dec. 9, 2022, in the Korean Intellectual Property Office, the disclosure of each of which is incorporated by reference herein in its entirety.

BACKGROUND
1. Field
[0002] The disclosure relates to an electronic device for transmitting media content and a method thereof.
2. Description of Related Art
[0003] Various services and additional functions provided through electronic devices, for example, portable electronic devices such as smart-phones, are gradually increasing. In order to increase the utility value of these electronic devices and satisfy the needs of various users, communication service providers or electronic device manufacturers are competitively developing electronic devices to provide various functions and differentiate themselves from other companies. Accordingly, various functions provided through electronic devices are becoming increasingly sophisticated.
[0004] The above information is presented as background information only to assist with an understanding of the disclosure. No determination has been made, and no assertion is made, as to whether any of the above might be applicable as prior art with regard to the disclosure.
SUMMARY
[0005] An electronic device may transmit and receive information to and from an external electronic device through a communication connection. When a communication connection with the electronic device is established, the external electronic device may transmit its own media content to a neighboring electronic device. The electronic device may need to, before transmitting media content selected for transmission, identify whether the external electronic device is to receive the media content. The external electronic device may be provided with a content preview to identify that the external electronic device is to receive the media content as a reception device. The content preview is preview content generated from the media content, and the generation and processing of the preview content may need to be performed according to the situation of a transmission device and a reception device. A neighboring external device may determine the quality, format, and amount of the preview content according to the communication status and support specifications of the external device before accepting the media content. A user of the reception device may identify whether to receive the media content by identifying the preview of

the media content from the transmission device.
[0006] Aspects of the disclosure are to address at least the above-mentioned problems and/or disadvantages and to provide at least the advantages described below. Accordingly, an aspect of the disclosure is to provide an electronic device for transmitting media content and a method thereof.
[0007] Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments.
[0008] In accordance with an aspect of the disclosure, an electronic device is provided. The electronic device includes a communication circuit, memory, including one or more storage media, storing instructions, and at least one processor communicatively coupled to the communication circuit and the memory, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to transmit, via the communication circuit, capability information of the electronic device and a request for transmitting capability information of an external electronic device to the external electronic device, obtain, via the communication circuit, the capability information of the external electronic device from the external electronic device, based on the obtained capability information of the external electronic device, generate preview content to be output from the external electronic device, and transmit, via the communication circuit, the generated preview content to the external electronic device.
[0009] The electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure is configured to, based on the discovery, provide capability information of the electronic device (the electronic device **101** of FIG. **1**) and request capability information of the external electronic device (the electronic device **102** of FIG. **1**) via the communication circuit (the communication module **190** of FIG. **2**).
[0010] The electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure is configured to, based on the request, receive capability information of the external electronic device (the electronic device **102** of FIG. **1**) from the external electronic device (the electronic device **102** of FIG. **1**) via the communication circuit (the communication module **190** of FIG. **2**).
[0011] The electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure is configured to, based on the received capability information of the external electronic device (the electronic device **102** of FIG. **1**), generate preview content associated with the media content and to be output from the external electronic device (the electronic device **102** of FIG. **1**).
[0012] The electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure is configured to transmit the generated preview content to the external electronic device (the electronic device **102** of FIG. **1**) via the communication circuit (the communication module **190** of FIG. **2**).
[0013] In accordance with another aspect of the disclosure, a method performed by an electronic device is provided. The method includes transmitting, via a communication circuit of the electronic device, capability information of the electronic device and a request for transmitting capability information of an external electronic device to the external electronic device, obtaining, via the communication circuit, the capability information of the external electronic device from the external electronic device, based on the obtained capability information of the external electronic device, generating preview content to be output from the external electronic device, and transmitting, via the communication circuit, the generated preview content to the external electronic device.
[0014] The method by an electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure includes, based on the discovery, providing capability information of the electronic device (the electronic device **101** of FIG. **1**) and requesting capability information of the external electronic device (the electronic device **102** of FIG. **1**).
[0015] The method by an electronic device (an electronic device **101** of FIG. **1**) according to an

embodiment of the disclosure includes, based on the request, receiving capability information of the external electronic device (the electronic device **102** of FIG. **1**) from the external electronic device (the electronic device **102** of FIG. **1**).
[0016] The method by an electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure includes, based on the received capability information of the external electronic device (the electronic device **102** of FIG. **1**), generating preview content associated with the media content and to be output from the external electronic device (the electronic device **102** of FIG. **1**).
[0017] The method by an electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure includes transmitting the generated preview content to the external electronic device (the electronic device **102** of FIG. **1**).
[0018] In accordance with another aspect of the disclosure, one or more non-transitory computer-readable storage media storing one or more computer programs including computer-executable instructions that, when executed by one or more processors of an electronic device individually or collectively, cause the electronic device to perform operations are provided. The operations include transmitting, via a communication circuit of the electronic device, capability information of the electronic device and a request for transmitting capability information of an external electronic device to the external electronic device, obtaining, via the communication circuit, the capability information of the external electronic device from the external electronic device, based on the obtained capability information of the external electronic device, generating preview content to be output from the external electronic device, and transmitting, via the communication circuit, the generated preview content to the external electronic device.
[0019] Other aspects, advantages, and salient features of the disclosure will become apparent to those skilled in the art from the following detailed description, which, taken in conjunction with the annexed drawings, discloses various embodiments of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0020] The above and other aspects, features, and advantages of certain embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:
[0021] FIG. **1** is a block diagram of an electronic device in a networked environment according to an embodiment of the disclosure;
[0022] FIG. **2** illustrates a block diagram of a preview module in an electronic device according to an embodiment of the disclosure;
[0023] FIG. **3** illustrates an operation of an electronic device according to an embodiment of the disclosure;
[0024] FIG. **4** illustrates an operation for transmitting preview content from a transmission device to a reception device according to an embodiment of the disclosure;
[0025] FIG. **5** illustrates an operation of adjusting the amount of preview content in an electronic device according to an embodiment of the disclosure;
[0026] FIG. **6**A is a diagram illustrating a function or operation in which an electronic device generates preview data of different quality according to the capabilities of each of a plurality of external electronic devices and transmits the generated preview data to each of the plurality of external electronic devices according to an embodiment of the disclosure;
[0027] FIG. **6**B is a diagram illustrating a function or operation in which an electronic device generates preview data, based on an electronic device having the lowest specification among the capabilities of each of a plurality of external electronic devices, and transmits the generated

preview data to each of the plurality of external electronic devices according to an embodiment of the disclosure; and

[0028] FIG. **6**C illustrates an operation performed according to each capability information for one or more external electronic devices in an electronic device according to an embodiment of the disclosure.

[0029] The same reference numerals are used to represent the same elements throughout the drawings.

DETAILED DESCRIPTION

[0030] The following description with reference to the accompanying drawings is provided to assist in a comprehensive understanding of various embodiments of the disclosure as defined by the claims and their equivalents. It includes various specific details to assist in that understanding but these are to be regarded as merely exemplary. Accordingly, those of ordinary skill in the art will recognize that various changes and modifications of the various embodiments described herein can be made without departing from the scope and spirit of the disclosure. In addition, descriptions of well-known functions and constructions may be omitted for clarity and conciseness.

[0031] The terms and words used in the following description and claims are not limited to the bibliographical meanings, but, are merely used by the inventor to enable a clear and consistent understanding of the disclosure. Accordingly, it should be apparent to those skilled in the art that the following description of various embodiments of the disclosure is provided for illustration purpose only and not for the purpose of limiting the disclosure as defined by the appended claims and their equivalents.

[0032] It is to be understood that the singular forms “a,” “an,” and “the” include a plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a component surface” includes reference to one or more of such surfaces.

[0033] As used herein, such an expression as “comprises” or “include” should not be interpreted to necessarily include all elements or all steps described in the specification, and should be interpreted to be allowed to exclude some of them or further include additional elements or steps.

[0034] As used herein, the terms including an ordinal number, such as expressions “a first” and “a second”, may be used to described various elements, but the corresponding elements should not be limited by such terms. The above terms are used merely for the purpose of distinguishing one element from other elements. For example, a first element may be termed a second element, and similarly, a second element may be termed a first element without departing from the scope of protection of the disclosure.

[0035] It should be understood that when an element is referred to as being “connected” or “coupled” to another element, it may be connected or coupled directly to the other element, or any other element may be interposer between them. Contrarily, in the case where an element is referred to as being “directly connected” or “directly coupled” to any other element, it should be understood that no other element exists therebetween.

[0036] Hereinafter, various embodiments according to the disclosure will be described in detail with reference to the accompanying drawings. Regardless of reference signs, the same or like elements are provided with the same or like reference signs in the drawings, and repetitive descriptions thereof will be omitted. In describing the disclosure below, a detailed description of known functions or configurations incorporated herein will be omitted when it is determined that the description may make the subject matter of the disclosure unnecessarily unclear. Furthermore, it should be noted that the accompanying drawings are presented merely to help easy understanding of the disclosure, and are not intended to limit the disclosure. The technical idea of the disclosure should be construed to cover all changes, equivalents, and alternatives, in addition to the accompanying drawings.

[0037] Hereinafter, a mobile station will be described with reference to drawings, but the mobile station may also be referred to as an electronic device, a terminal, a mobile equipment (ME), a user equipment (UE), a user terminal (UT), a subscriber station (SS), a wireless device, a handheld device, and an access terminal (AT). The mobile station may also be a device with communication capabilities, such as a cell phone, a personal digital assistant (PDA), a smart-phone, a wireless modem, and a laptop.

[0038] It should be appreciated that the blocks in each flowchart and combinations of the flowcharts may be performed by one or more computer programs which include instructions. The entirety of the one or more computer programs may be stored in a single memory device or the one or more computer programs may be divided with different portions stored in different multiple memory devices.

[0039] Any of the functions or operations described herein can be processed by one processor or a combination of processors. The one processor or the combination of processors is circuitry performing processing and includes circuitry like an application processor (AP, e.g. a central processing unit (CPU)), a communication processor (CP, e.g., a modem), a graphics processing unit (GPU), a neural processing unit (NPU) (e.g., an artificial intelligence (AI) chip), a wireless fidelity (Wi-Fi) chip, a Bluetooth® chip, a global positioning system (GPS) chip, a near field communication (NFC) chip, connectivity chips, a sensor controller, a touch controller, a finger-print sensor controller, a display driver integrated circuit (IC), an audio CODEC chip, a universal serial bus (USB) controller, a camera controller, an image processing IC, a microprocessor unit (MPU), a system on chip (SoC), an IC, or the like.

[0040] FIG. **1** is a block diagram illustrating an electronic device **101** in a network environment **100** according to an embodiment of the disclosure.

[0041] Referring to FIG. **1**, the electronic device **101** in the network environment **100** may communicate with an electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or at least one of an electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). According to an embodiment, the electronic device **101** may communicate with the electronic device **104** via the server **108**. According to an embodiment, the electronic device **101** may include a processor **120**, memory **130**, an input device **150**, a sound output device **155**, a display device **160**, an audio module **170**, a sensor module **176**, an interface **177**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In some embodiments, at least one of the components (e.g., the display device **160** or the camera module **180**) may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. In some embodiments, some of the components may be integrated into a single element. For example, the sensor module **176** (e.g., a fingerprint sensor, an iris sensor, or an illumination sensor) may be embedded and implemented in the display device **160**.

[0042] The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation. According to one embodiment, as at least part of the data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in non-volatile memory **134**. According to an embodiment, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. Additionally or alternatively, the auxiliary processor **123** may be configured to use lower power than the main processor **121** or be specific to a designated function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.

[0043] The auxiliary processor **123** may control at least some of functions or states related to at least one component (e.g., the display device **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application).
[0044] The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**. The non-volatile memory **134** may include internal memory **136** or external memory **138**.
[0045] The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.
[0046] The input device **150** may receive a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input device **150** may include, for example, a microphone, a mouse, a keyboard, or a digital pen (e.g., a stylus pen).
[0047] The sound output device **155** may output sound signals to the outside of the electronic device **101**. The sound output device **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.
[0048] The display device **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display device **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display device **160** may include a touch circuitry adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.
[0049] The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input device **150**, or output the sound via the sound output device **155** or a headphone of an external electronic device (e.g., an electronic device **102**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **101**.
[0050] The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.
[0051] The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly (e.g., wiredly) or wirelessly. According to an embodiment, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.
[0052] A connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**). According to an embodiment, the connecting terminal **178** may include, for example, a HDMI connector, a USB connector, a SD card connector, or an audio connector (e.g., a headphone connector).
[0053] The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a

vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

[0054] The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

[0055] The power management module **188** may manage power supplied to the electronic device **101**. According to one embodiment, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

[0056] The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

[0057] The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a cellular network, the Internet, or a computer network (e.g., LAN or wide area network (WAN)). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify and authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.

[0058] The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. According to an embodiment, the antenna module may include an antenna including a radiating element composed of a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna module **197** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** (e.g., the wireless communication module **192**) from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**.

[0059] At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

[0060] According to an embodiment, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the electronic devices **102** or **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to an embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102**, **104**, or server **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, or client-server computing technology may be used, for example.

[0061] FIG. **2** illustrates a block diagram of a preview module in an electronic device according to an embodiment of the disclosure. The functions or operations performed by the preview module according to an embodiment of the disclosure may be performed by the processor **120**, and the preview module may be included as a configuration element of the processor **120**.

[0062] According to an embodiment of the disclosure, the electronic device **101** may include a preview module **220**. In the electronic device **101** of the disclosure, the preview module **220** may be configured to connect the electronic device **101** (e.g., a transmission device) with the external electronic device **102** or **104** (e.g., a reception device) to identify the performance of a transmission device **101** and generate an appropriate video preview file accordingly.

[0063] Referring to FIG. **4**, the preview module **220** according to an embodiment of the disclosure may include a connection module **401**. In an example, the connection module **401** may be configured to perform wireless communication for transmission and reception of preview content between electronic devices. In an example, the connection module **401** may discover and connect an electronic device, based on a wireless communication technology such as Wi-Fi Direct or Wi-Fi Aware. In an example, the electronic device **101** may use UPnP as a method to discover the external electronic device **102** or **104**.

[0064] Referring to FIG. **4**, the preview module **220** according to an embodiment of the disclosure may include a capability module **403**. Capability information and/or preview data according to an embodiment of the disclosure may be stored at least temporarily in memory **130**. In an example, capability information may be exchanged between the electronic device **101** and the external electronic device **102** or **104** and stored. In an example, the electronic device **101** may transmit its own unique capability information to the external electronic device **102** or **104** to generate and transmit a content preview to the external electronic device **102** or **104**, and may request capability information from the external electronic device **102** or **104** required to generate and transmit the content preview. In an example, the capability information of the electronic device **101** and the external electronic device **102** or **104** may include at least one of remaining storage space of a reception device **102**, chipset information, supported codecs, resolution, and frames per second (fps). In an example, the external electronic device **102** or **104** may respond to a request from the electronic device **101**, and store, in memory (e.g., memory **130** of FIG. **3**) of the electronic device **101**, the capability information exchanged between the devices by including its own capability information. In an example, in a first connection between the electronic device **101** and the external electronic device **102** or **104**, unique capability information for each of the electronic device **101** and the external electronic device **102** or **104** is exchanged and stored, and thus unnecessary information exchange procedures for subsequent connections may be prevented from being duplicated. In an example, the electronic device **101** may determine preview content to be transmitted, according to the capability of the external electronic device **102** or **104**.

[0065] Referring to FIG. **2**, the preview module **220** according to an embodiment of the disclosure may include a create, read, update, and delete (CRUD) module **405**. The CRUD module may be configured to generate and update preview content. Although shown as separate modules of FIG. **4**, they are shown as examples for illustrative purposes only and may not be limited to operations performed by a single module. The operations described in the disclosure may be performed by at least one processor. A preview module according to an embodiment of this document may be implemented as a software module executable by at least one processor.

[0066] The electronic device **101** according to an example of the disclosure may perform at least one of the operations of generating (creating), reading, storing, updating, or deleting preview content by means of the CRUD module **405**.

[0067] In an example, in the electronic device **101**, with respect to the operation of generating the preview content, in case that the preview content of the video to be transmitted to the external electronic device **102** or **104** is more over spec than the capability of the external electronic device **102** or **104**, and at the same time, memory **130** (e.g., a database (video preview DB)) of the preview content does not have a video preview that is substantially the same as the capability of the external electronic device **102** or **104** or that is more under spec than the capability thereof, the video preview content for the video to be transmitted may be generated. The electronic device **101** (e.g., the transmission device) according to an embodiment of the disclosure may generate preview data, based on the state of the external electronic device **102** or **104** (e.g., if the remaining capacity of the external electronic device **102** or **104** is equal to or less than a designated capacity value) and/or the state of the network to which the electronic device **101** is connected.

[0068] In an example, the electronic device **101** may generate preview content, based on the capability information of the reception device **102** and the capability information of the transmission device **101**, which are received by the electronic device **101**. In an example, the electronic device **101** may convert the preview content into an optimized version by considering at least one of a codec, frames per second (fps), and resolution supported by the external electronic device **102** or **104** via the capability information of the external electronic device **102** or **104**. However, depending on at least one of the remaining storage space of the external electronic device **102** or **104** and the performance of the chipset, the preview content may be converted to a lower specification than the optimized version.

[0069] In an example, in the electronic device **101**, the generated preview content may be stored in memory **130** (e.g., a database) for later reuse. For example, when the electronic device **101** generates preview content about the video content to be transmitted, based on information from the external electronic device **102** or **104**, the electronic device **101** may use previously generated preview content stored in memory **130** without having to generate the preview content again in the future.

[0070] The electronic device **101** according to an example of the disclosure may perform an operation of reading preview content. In an example, before transmitting a video preview, the transmission device **101** may first search a database to see if there is video preview content that meets or falls below the support specification according to the capability information of the external electronic device **102** or **104** before transmitting the video preview. In an example, since the electronic device **101** may transmit the preview content before transmitting the original file of the video preview content, if the specification of the generated preview content is lower than the support specification of the external electronic device **102** or **104**, the corresponding preview content may be transmitted.

[0071] The electronic device **101** according to an example of the disclosure may update or delete preview content. In an example, when there is preview content stored in a database with respect to video content to be transmitted, if preview content generated with a different specification is generated for the same video content, the electronic device **101** may delete pre-existing preview content. In an example, the electronic device **101** may delete preview content so that multiple pieces of preview content do not exist for the video content to be transmitted, thereby efficiently using storage capacity. In this case, the electronic device **101** may also contribute to speed improvements because the electronic device **101** may not need to perform operations of searching for or selecting preview content when multiple pieces of preview content exist for the same video content.

[0072] The electronic device **101** according to an example of the disclosure may transmit preview content. In an example, the electronic device **101** may receive a request from the external electronic device **102** or **104** and/or an application running on the electronic device **101** to transmit video preview content. In this case, the electronic device **101** may transmit the preview content in response to the request. In an example, the electronic device **101** may transmit the preview content to the external electronic device **102** or **104** via a transmission module **407** within the video preview module **220**. In an example, the transmission of the original video content to be transmitted by the electronic device **101** and the transmission of the content preview generated for the video to be transmitted may be processed in separate layers. In this case, the electronic device **101** may transmit the preview content without affecting the transmission performance of the original video content.

[0073] FIG. **3** illustrates an operation of an electronic device according to an embodiment of the disclosure. The electronic device **101** according to an embodiment of the disclosure may determine preview content that may be transmitted according to the capability of the external electronic device **102** or **104**.

[0074] Referring to FIG. **3**, in operation **501**, the electronic device **101** may identify whether the external electronic device **102** or **104** has sufficient capability for the video preview, based on the capability information received from the external electronic device **102** or **104**. In an example, the identifying whether the external electronic device **102** or **104** has sufficient capability for video preview may be an operation of identifying whether the electronic device **101** is capable of receiving the generated preview content from the external electronic device **102** or **104** and displaying the same. In an example, the electronic device **101** may identify whether the quality of the generated preview content is maintainable in the specification supported by the external electronic device **102** or **104**. In an example, the capability information of the reception device **102** may include at least a part of the information related to codec, fps, and resolution specifications.

[0075] In FIG. **3**, when it is identified that the external electronic device **102** or **104** has sufficient capability to receive the preview content ("yes" in operation **501**), the electronic device **101** may transmit all pieces of preview of each piece of video content to be transmitted, in operation **503**. In an example, the electronic device **101** may select one or more pieces of video content to be transmitted, and may transmit one or more pieces of preview content in the selected video content. Here, the electronic device **101** may transmit all pieces of preview content to be transmitted in case that the electronic device **101** identifies, based on the capability information of the external electronic device **102** or **104**, that the one or more pieces of preview content are receivable by the external electronic device **102** or **104**.

[0076] In FIG. **3**, when it is identified that the external electronic device **102** or **104** has insufficient capability to receive the preview content ('no" in operation **501**), the electronic device **101** may identify whether the external electronic device **102** or **104** has sufficient reception performance in operation **505**. For example, the electronic device **101** may identify at least a part of the network communication status, remaining storage space, and chipset information of the external electronic device **102** or **104**. Alternatively, the electronic device **101** may identify that the quality of the preview content to be transmitted from the electronic device **101** can be maintained in the external electronic device **102** or **104**. For example, it may be identified whether at least one of the specifications of the codec, fps, and/or resolution for the preview content of the electronic device

101 to be compatible on the external electronic device **102** or **104** satisfies a preview quality maintenance condition. For example, the electronic device **101** according to an embodiment of the disclosure may transmit the preview content to the external electronic device **102** or **104** in case that all of the codec, fps, and resolution for the preview content to be compatible on the external electronic device **102** or **104** satisfy the preview quality maintenance condition (e.g., whether the preview content can be output via the external electronic device **102** or **104**). In other words, in case that any one of the specifications of the codec, fps, and resolution for the preview content to be compatible on the external electronic device **102** or **104** does not satisfy the quality maintenance condition, the quality of the preview content may be degraded and transmitted to the external electronic device **102** or **104**.

[0077] In FIG. **3**, when it is identified that the reception performance of the external electronic device **102** or **104** is sufficient ("yes" in operation **505**), the electronic device **101** may transcode the preview for each piece of video content to match the capability of the external electronic device **102** or **104** and then transmit the transcoded preview in operation **507**.

[0078] In FIG. **3**, when it is identified that the reception performance of the external electronic device **102** or **104** is not sufficient ("no" in operation **505**), the electronic device **101** may identify whether there is one piece of video content to be transmitted in operation **509**. In FIG. **3**, when one piece of video content is to be transmitted ("yes" in operation **509**), the electronic device **101** may transmit a thumbnail of video content to the external electronic device **102** or **104** in operation **511**. In FIG. **3**, when more than one piece of video content is to be transmitted ("no" in operation **509**), the electronic device **101** may generate a designated file format (e.g., GIF file) using a thumbnail of each piece of video content and transmit the same in operation **513**.

[0079] FIG. **4** illustrates operations for transmitting preview content from a transmission device to a reception device according to an embodiment of the disclosure.

[0080] Referring to FIG. **4**, in operation **601**, a preview module **220***a* according to an example of the disclosure may transmit preview content and receive a request for preview content from an application (or, a service). In an example, the electronic device **101** may select video content to be transmitted by a user and execute a service to share the same with an external electronic device **102** or **104**. In an example, when a service or application of the transmission device (e.g., electronic device **101**) requests a video preview, the preview module **220***a* (e.g., the processor **120** of FIG. **1**) may request connection to the external electronic device **102** or **104**. In an example, the electronic device **101** may receive network information of the external electronic device **102** or **104** and perform a connection request to the external electronic device **102** or **104**, based on the network information. In an example, the connection request may be based on a wireless communication technology such as, but not limited to, Wi-Fi Direct or Wi-Fi Aware. For example, UPnP or the like may additionally be used to find a reception device.

[0081] In FIG. **4**, the electronic device **101** may perform a discovery operation to connect with the external electronic device **102** or **104** in operation **602**. In this case, the electronic device **101** may identify whether services and applications associated with reception of preview content exist on the external electronic device **102** or **104**.

[0082] In FIG. **4**, the electronic device **101** may request capability information from the reception device, in order to receive capability information from the external electronic device **102** or **104**, in operation **603**. Further, the electronic device **101** may notify the external electronic device **102** or **104** of its own capability information.

[0083] In FIG. **4**, the electronic device **101** may receive capability information from the external electronic device **102** or **104** in operation **604**. In an example, when the preview module **220***a* of the electronic device **101** is connected to the preview module **220***b* of the external electronic device **102** or **104**, they may exchange capability information to identify whether video preview is supported, whether a codec is supported, and/or whether a resolution is supported. In an example, based on the capability information from the external electronic device **102** or **104**, the electronic device **101** may reuse stored preview content or generate new preview content to transmit multiple pieces of video preview content corresponding to the requested service or application. In an example, a service or application of the electronic device **101** may provide a preview of video content according to the performance of the external electronic device **102** or **104** by requesting the transmission of content to be transmitted together with preview content from a service or application node of the external electronic device **102** or **104** (indicated by reference numeral **608**).

[0084] According to an embodiment of the disclosure, the external electronic device **102** or **104** may provide the transmission device **101** with an option for the video preview content to be received. In an example, the external electronic device **102** or **104** may notify the electronic device **101** of options, such as whether reception of the video preview content is possible, the format of the preview content (image format or video format, such as JPEG, GIF, etc.) optimized for reception by the external electronic device **102** or **104**, and/or the quality of the preview content. In addition, at a time point when the external electronic device **102** or **104** exchanges capability information for preview content with the electronic device **101**, the preview content capability information may be delivered together with information about options for preview content optimized for reception by the external electronic device **102** or **104** and may be used to determine preview content to be transmitted by the transmission device. In an example, the external electronic device **102** or **104** may store the received preview content. Further, the external electronic device **102** or **104** may bundle and store the received preview content in separate albums. In an example, values for the corresponding option (e.g., an option regarding whether preview content is received and/or the quality of the preview content, etc.) may be exchanged together when the transmission/reception device exchanges capability information with each other, so that only the original content to be transmitted may be received without delivery of additional preview content. Alternatively, a user-selectable user interface (UI) may be provided to enable the preview content to be provided in a format desired by the transmission/reception device.

[0085] In FIG. **4**, in operation **605**, the electronic device **101** may, based on the capability information received from the external electronic device **102** or **104**, transmit media content to the external electronic device **102** or **104** without generating preview content in case that the electronic device **101** is unable to display the preview content, such as the case in which the external electronic device **102** or **104** does not have the preview module **220**, or in which the capability request from the external electronic device **102** or **104** exceeds a timeout. Alternatively, the electronic device **101** may not provide a preview for video content among the selected content, or may only provide a preview for image content.

[0086] In FIG. **4**, the electronic device **101** may generate preview content in the preview module **220***a* in operation **606**. For example, the generation of the preview content may comply with the description of FIGS. **4** and **3**.

[0087] In FIG. **4**, in operation **607**, the preview content generated by the preview module **220***a* of the electronic device **101** may be delivered to the application node for transmission to the external electronic device **102** or **104**, or the preview content may be transmitted from the preview module of the electronic device **101** to the preview module of the external electronic device **102** or **104** and the resulting value may be delivered to the application node.

[0088] In FIG. **4**, in operation **608**, the electronic device **101** may transmit the generated preview content to the external electronic device **102** or **104** to request reception of the media content (indicated by reference numeral **608**).

[0089] FIG. **5** illustrates an operation of adjusting the amount of preview content in an electronic device according to an embodiment of the disclosure.

[0090] According to an embodiment of the disclosure, the electronic device **101** includes a preview module **220**, and the preview module **220** may be executed at the request of a service or application of the electronic device **101** that transmits media content. In an example, the network information of the external electronic device **102** or **104** delivered together at the request of the electronic device **101** may be used to connect the preview modules **220***a* and **220***b* of the electronic device **101** and the external electronic device **102** or **104**, respectively. For example, the electronic device **101** may identify whether the capability information of the external electronic device **102** or **104** is stored in a database, and if so, the electronic device **101** may not request the capability information. In an example, the capability information exchanged between the transmission device **101** and the external electronic device **102** or **104** may include the range of support for the codec, chipset information, whether HDR10+ is supported, etc. In an example, each of the transmission device **101** and the external electronic device **102** or **104** may store the capability information of the other device in an internal database so that they do not perform the same operation repeatedly after the initial connection.

[0091] Referring to FIG. **5**, the electronic device **101** may identify whether the remaining capacity of the external electronic device **102** or **104** is sufficient in operation **701**. For example, the electronic device **101** may identify whether the external electronic device **102** or **104** has free capacity to display the preview content to be transmitted to the external electronic device **102** or **104**.

[0092] In FIG. **5**, when the remaining capacity of the external electronic device **102** or **104** is insufficient ("no" in operation **701**), the electronic device **101** may adjust the amount of preview content to be transmitted in operation **703**. For example, when there is more than one piece of video preview content, the electronic device **101** may adjust one piece of video preview content to be transmitted. In this case, the electronic device **101** may adjust the amount of preview content to be transmitted by selecting an appropriate preview content from among multiple pieces of video preview content according to the remaining capacity of the external electronic device **102** or **104**.

[0093] In FIG. **5**, when the remaining capacity of the external electronic device **102** or **104** is sufficient ("yes" in operation **701**) or the amount of preview content to be transmitted is adjusted although the remaining capacity is not sufficient (operation **703**), the electronic device **101** may identify whether the reception performance of the external electronic device **102** or **104** is sufficient, based on the state of a network to which the transmission/reception device is connected, chipset information of the reception device, etc., in operation **705**.

[0094] In FIG. **5**, when it is identified that the reception performance of the external electronic device **102** or **104** is insufficient, the electronic device **101** may adjust the amount of preview content to be transmitted in operation **707**. In an example, the receivable amount of preview content may be adjusted according to the reception performance, based on the capability information of each of the electronic device **101** and the external electronic device **102** or **104**. In addition, the electronic device **101** may adjust the amount of preview content to be transmitted, according to the network communication status of the external electronic device **102** or **104**.

[0095] In FIG. **5**, when it is identified that the reception performance of the external electronic device **102** or **104** is sufficient ("yes" in operation **705**) or when the amount of preview content to be transmitted is adjusted ("yes" in operation **707**), the electronic device **101** may identify whether the preview content to be transmitted falls within the range of support for a receiving codec in operation **709**. For example, the electronic device **101** may identify whether the codec of the preview content generated by the electronic device **101** is a codec supported by the external electronic device **102** or **104**, based on the capability information of each of the electronic device **101** and the external electronic device **102** or **104**.

[0096] In FIG. **5**, when the codec of the preview content generated by the electronic device **101** is not a codec supported by the external electronic device **102** or **104** ("no" in operation **709**), the electronic device **101** may process the preview content by performing a transcoding or conversion operation on the preview content in operation **711**.

[0097] In FIG. **5**, when the preview content to be transmitted falls within the range of support for a receiving codec of the external electronic device **102** or **104** ("yes" in operation **709**) or when the preview content has been processed to fall within the codec support range of the external electronic device **102** or **104** ("yes" in operation **711**), the electronic device **101** may identify whether the technology used for the preview content to be transmitted is within the codec support range of the external electronic device **102** or **104** in operation **713**. In an example, the electronic device **101** may, based on capability information of each of the electronic device **101** and the external electronic device **102** or **104**, identify whether the technology used by the electronic device **101** in the process of generating the preview content is within the range supported by the external electronic device **102** or **104**.

[0098] In FIG. **5**, in operation **715**, the electronic device **101** may perform an operation of converting the preview content to process the same into preview content within the range supported by the external electronic device **102** or **104**.

[0099] According to the operation of the electronic device **101** in accordance with an embodiment of the disclosure, the electronic device **101** may identify capability information of the external electronic device **102** or **104**, and generate and transmit preview content based on the capability information. In other words, by incrementally and/or dynamically generating and processing the preview content, the electronic device **101** may provide a preview so that the external electronic device **102** or **104** may identify information about the content to be received by the external electronic device **102** or **104** in operation before transmitting the original content, regardless of the performance of the electronic device **101** or the unique characteristics of the content.

[0100] Referring to FIGS. **6**A, **6**B, and **6**C, it is illustrated that in an electronic device **101** in accordance with an embodiment of the disclosure, when connected to a plurality of external electronic devices **102** and **104**, the resolutions supported by each of the reception devices are different. It is also illustrated that for each of the plurality of reception devices, the video preview content may be omitted or provided in an image format (e.g., jpeg, gif) depending on storage space, chipset information, CPU usage, and/or options configured by the reception device.

[0101] FIG. **6**A is a diagram illustrating a function or operation in which an electronic device generates preview data of different quality based on capabilities of each of the plurality of external electronic devices and transmits the generated preview data to each of the plurality of external electronic devices according to an embodiment of the disclosure.

[0102] FIG. **6**B is a diagram illustrating a function or operation in which an electronic device generates preview data based on an electronic device having the lowest specification among the capabilities of each of the plurality of external electronic devices, and transmits the generated preview data to each of the plurality of external electronic devices according to an embodiment of the disclosure.

[0103] FIG. **6**C illustrates an operation performed according to each capability information for one or more external electronic devices in an electronic device according to an embodiment of the disclosure.

[0104] In an example, a transmission device **301***b* is an electronic device **101** capable of supporting ULTRA HD of 8K, and the transmission device **301***b* may share video content with a first reception device **303***a* and a second reception device **303***b*. Here, the first reception device **303***a* and the second reception device **303***b* may be a device supporting ULTRA HD of 8K and a device supporting FULL HD of 1080P, respectively. For example, the quality of the preview content may be differentiated based on the performance of the plurality of reception devices **303***a* and **303***b* having different specifications. In an example, a scenario of sharing a video in a situation where the transmission device and the reception device is 1:N, such as the transmission device **301***b* versus the reception devices **303***a* and **303***b*, may be as follows. For example, referring to FIG. **6**A, in the case of transmitting different video preview content to the N reception devices, respectively, the capability information of each of the N reception devices may be identified and preview content suitable for each of the N reception devices may be transmitted. However, in order to minimize the inefficiency of generating N pieces of video preview content, the transmission device may identify the capability information of the N reception devices, generate one video preview content based on the reception device having the lowest specification, and transmit the one video preview in batches. Referring to FIG. **6**B, a relatively high-specification terminal among N reception devices may receive video preview content of 8K quality in a 1:1 transmission scenario, but may transmit video preview content of FULL HD specification, based on the lowest specification in a 1:N transmission scenario. For example, in FIG. **6**B, a transmission device **301***b* may produce video preview content of 8K quality and receive capability information from each of a first reception device **303***a* and a second reception device **303***b*. In this case, the transmission device **301***b* may generate video preview content of FULL HD specification, according to the second reception device **303***b* providing a relatively lower specification among the plurality of reception devices **303***a* and **303***b*. The transmission device **301***b* may transmit the video preview content of FULL HD specification to the first reception device **303***a* and the second reception device **303***b*. The electronic device **101** (e.g., the transmission device) according to an embodiment of the disclosure may identify whether the external electronic device **102** or **104** (e.g., the reception device) supports HDR. The electronic device **101** according to an embodiment of the disclosure may, in case that the external electronic device **102** or **104** is identified as having substantially the same HDR capability as the HDR capability of the electronic device **101**, transmit preview content based on the HDR capability of the electronic device **101** to the external electronic device **102** or **104**. The electronic device **101** according to an embodiment of the disclosure may, in case that the external electronic device **102** or **104** is identified as having a capability that is not substantially the same as (e.g., lower than) the HDR capability of the electronic device **101**, generate preview content based on the HDR capability of the external electronic device **101** and transmit the generated preview content to the external electronic device **102** or **104**. To this end, the electronic device **101** according to an embodiment of the disclosure may perform tone mapping, perform gamut mapping, or perform format conversion on the preview content generated according to the HDR capability of the electronic device **101**. Gamut mapping according to an embodiment of the disclosure may refer to a function or operation of reducing the color gamut of content from DCI-P3 area content (wide-range color expression capability) to rec.709 (narrow-range color expression capability). Format conversion according to an embodiment of the disclosure may refer to a function or operation of converting HDR10+ to HDR10, or converting HDR10 to HLG. The electronic device **101** according to an embodiment of the disclosure may, when the external electronic device **102** or **104** only supports SDR, also convert HDR content to SDR content and transmit the SDR content to the external electronic device **102** or **104**.

[0105] FIG. **6**C illustrates an operation performed according to each capability information for one or more external electronic devices in an electronic device according to an embodiment of the disclosure.

[0106] In an example, when a transmission device and a reception device are in a 1:N relationship, the preview quality may be unified according to the performance of the reception devices when sharing video content. In an example, a transmission device **301***b* may request capability information for preview content from N reception devices. For example, the transmission device **301***b* may request capability information for the preview content from each of the N reception devices and receive all N responses, which may take a considerable amount of time. Therefore, the transmission device **301***b* may arrange a time (timer) long enough to have no influence on the transmission performance of the original media content to be transmitted, and may not support preview content for the reception devices from which no respond is received within the arranged time. Referring to FIG. **6**C, in case that one of the three reception devices **301***a*, **301***b*, and **303***c* fails to receive a response to the request for capability information within a predetermined time, the transmission device **301***b* may determine that it is not possible to identify the capability for the reception device (e.g., **303***c*) and may not provide the video preview content.

[0107] According to an embodiment of the disclosure, the electronic device **101** and the external electronic device **102** or **104** may exchange capability information with each other. In an example, the electronic device **101** may provide optimized video preview content to the external electronic device **102** or **104**. In an example, the electronic device **101** may provide the preview content to the external electronic device **102** or **104** so that the external electronic device **102** or **104** may identify information about the video content to be transmitted in operation prior to accepting or transmitting a file from the external electronic device **102** or **104**.

[0108] An electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure may include a communication circuit (a communication module **190** of FIG. **1**), and at least one processor (**120** of FIG. **1**), wherein the at least one processor is configured to discover an external electronic device (**102** of FIG. **1**) for transmission of media content from the electronic device (the electronic device **101** of FIG. **1**).

[0109] An electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure may be configured to provide capability information of the electronic device (the electronic device **101** of FIG. **1**) via the communication circuit (**190** of FIG. **2**) based on the discovery, and to request capability information of the external electronic device (**102** of FIG. **1**).

[0110] An electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure may be configured to, based on the request, receive capability information of the external electronic device (**102** of FIG. **1**) from the external electronic device (**102** of FIG. **1**) via the communication circuit (**190** of FIG. **2**).

[0111] An electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure may be configured to generate preview content associated with the media content and to be output from the external electronic device (**102** of FIG. **1**), based on the received capability information of the external electronic device (**102** of FIG. **1**).

[0112] An electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure may be configured to transmit the generated preview content to the external electronic device (the electronic device **102** of FIG. **1**) via the communication circuit (**190** of FIG. **2**).

[0113] An electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure may be further configured to identify whether the generated preview content exists for the media content.

[0114] An electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure may be further configured to, in case that the generated preview content exists for the media content, process the generated preview content based on the capability information of the external electronic device (**102** of FIG. **1**).

[0115] In an electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure, the capability information received from the external electronic device (**102** of FIG. **1**) may include information related to at least one of a codec, frames per second (FPS), resolution, chipset, storage capacity, and network communication, of the external electronic device (**102** of FIG. **1**).

[0116] In an electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure, the capability information received from the external electronic device (**102** of FIG. **1**) may include information related to the quality of the preview content that the external electronic device (**102** of FIG. **1**) is capable of receiving.

[0117] An electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure may be configured to adjust the amount of the preview content to be transmitted to the external electronic device (**102** of FIG. **1**), based on the received capability information of the external electronic device (**102** of FIG. **1**).

[0118] In an electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure, the media content includes video content and image content, and the electronic device may be further configured to reduce the amount of the preview content for the video content to be transmitted to the external electronic device (**102** of FIG. **1**) in case that the remaining capacity of the external electronic device (**102** of FIG. **1**) is equal to or less than a predetermined amount of capacity, or that the reception performance of the external electronic device (**102** of FIG. **1**) is not capable of maintaining the quality of the generated preview content.

[0119] An electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure may be configured to, in case that the amount of preview content to be transmitted is adjusted, identify whether the preview content to be transmitted is capable of maintaining a predetermined quality on the external electronic device (**102** of FIG. **1**).

[0120] An electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure may be configured to, in case that the preview content to be transmitted is capable of maintaining the predetermined quality on the external electronic device (**102** of FIG. **1**), transmit the preview content to be transmitted to the external electronic device (**102** of FIG. **1**) via the communication circuit.

[0121] An electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure may be further configured to, in case that the preview content to be transmitted is unable to maintain the predetermined quality on the external electronic device (**102** of FIG. **1**), process the preview content to be transmitted, in a format supportable by the external electronic device (**102** of FIG. **1**).

[0122] In an electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure, the external electronic device (**102** of FIG. **1**) includes a plurality of external electronic devices **102** and **104**, and the at least one processor may be configured to receive capability information for each of the plurality of reception devices (**303***a*, **303***b*, and **303***c* of FIG. **6**C).

[0123] An electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure may be configured to identify the media content to be transmitted to the plurality of reception devices (**303***a*, **303***b*, and **303***c* of FIG. **6**C).

[0124] An electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure may be configured to, in case that the media content to be transmitted is the same, produce the preview content based on a reception device having the lowest specification among the plurality of reception devices (**303***a*, **303***b*, and **303***c* of FIG. **6**C).

[0125] An electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure may be configured to receive the capability information for each of the plurality of reception devices (**303***a*, **303***b*, and **303***c* of FIG. **6**C) within a predetermined time, and not to transmit the preview content to the corresponding reception device in case that the capability information is received after the predetermined time.

[0126] A method by an electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure may include discovering an external electronic device (**102** of FIG. **1**) for transmission of media content from the electronic device (the electronic device **101** of FIG. **1**).

[0127] A method by an electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure may include, based on the discovery, providing capability information of the electronic device (the electronic device **101** of FIG. **1**) and requesting capability information of an external electronic device (an electronic device **102** of FIG. **1**).

[0128] A method by an electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure may include, based on the request, receiving capability information of the external electronic device (**102** of FIG. **1**) from the external electronic device (**102** of FIG. **1**).

[0129] A method by an electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure may include, based on the received capability information of an external electronic device (**102** of FIG. **1**), generating preview content associated with the media content and to be output from the external electronic device (**102** of FIG. **1**).

[0130] A method by an electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure may include transmitting the generated preview content to the external electronic device (**102** of FIG. **1**).

[0131] A method by an electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure may further include identifying whether the generated preview content exists for the media content.

[0132] A method by an electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure may further include, in case that the generated preview content exists for the media content, processing the generated preview content for the media content, based on the capability information of the external electronic device (**102** of FIG. **1**).

[0133] A method by an electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure, the capability information received from the external electronic device (**102** of FIG. **1**) may include at least one of information related to a codec, frames per second (FPS), resolution, chipset, storage capacity, and network communication, of the external electronic device (**102** of FIG. **1**).

[0134] In a method by an electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure, the capability information received from the external electronic device (**102** of FIG. **1**) may include information related to the quality of the preview content that the external electronic device (**102** of FIG. **1**) is capable of receiving.

[0135] A method by an electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure may include, based on the received capability information of the external electronic device (**102** of FIG. **1**), adjusting the amount of the preview content to be transmitted to the external electronic device (**102** of FIG. **1**).

[0136] In a method by an electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure, the media content includes video content and image content, and the method may further include reducing the amount of the preview content for the video content to be transmitted to the external electronic device (**102** of FIG. **1**) in case that the remaining capacity of the external electronic device (**102** of FIG. **1**) is equal to or less than a predetermined amount of capacity, or that the reception performance of the external electronic device (**102** of FIG. **1**) is not capable of maintaining the quality of the generated preview content.

[0137] A method by an electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure may include, in case that the amount of preview content to be transmitted is adjusted, identifying whether the preview content to be transmitted is capable of maintaining a predetermined quality on the external electronic device (**102** of FIG. **1**).

[0138] A method by an electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure may include, in case that the preview content to be transmitted is capable of maintaining the predetermined quality, transmitting the preview content to be transmitted to the external electronic device (**102** of FIG. **1**) via the communication circuit.

[0139] A method by an electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure may further include, in case that the preview content to be transmitted is unable to maintain the predetermined quality, processing the preview content to be transmitted, in a format supportable by the external electronic device (**102** of FIG. **1**).

[0140] In a method by an electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure, the external electronic device (**102** of FIG. **1**) includes a plurality of external electronic devices **102** and **104**, and the method may further include receiving capability information for each of the plurality of reception devices (**303***a*, **303***b*, and **303***c* of FIG. **6**C).

[0141] A method by an electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure may further include identifying the media content to be transmitted to the plurality of reception devices (**303***a*, **303***b*, and **303***c* of FIG. **6**C).

[0142] A method by an electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure may further include, in case that the media content to be transmitted

is the same, producing the preview content based on a reception device having the lowest specification among the received capabilities of the plurality of external electronic devices (**303***a*, **303***b*, and **303***c* of FIG. **6**C).
[0143] In a method by an electronic device (an electronic device **101** of FIG. **1**) according to an embodiment of the disclosure, in relation to an operation of receiving the capability information for each of the plurality of reception devices (**303***a*, **303***b*, and **303***c* of FIG. **6**C) within a predetermined time, the preview content may not be transmitted to the corresponding reception device in case that the capability information is received after the predetermined time.
[0144] It will be appreciated that various embodiments of the disclosure according to the claims and description in the specification can be realized in the form of hardware, software or a combination of hardware and software.
[0145] Any such software may be stored in non-transitory computer readable storage media. The non-transitory computer readable storage media store one or more computer programs (software modules), the one or more computer programs include computer-executable instructions that, when executed by one or more processors of an electronic device individually or collectively, cause the electronic device to perform a method of the disclosure.
[0146] Any such software may be stored in the form of volatile or non-volatile storage such as, for example, a storage device like read only memory (ROM), whether erasable or rewritable or not, or in the form of memory such as, for example, random access memory (RAM), memory chips, device or integrated circuits or on an optically or magnetically readable medium such as, for example, a compact disk (CD), digital versatile disc (DVD), magnetic disk or magnetic tape or the like. It will be appreciated that the storage devices and storage media are various embodiments of non-transitory machine-readable storage that are suitable for storing a computer program or computer programs comprising instructions that, when executed, implement various embodiments of the disclosure. Accordingly, various embodiments provide a program comprising code for implementing apparatus or a method as claimed in any one of the claims of this specification and a non-transitory machine-readable storage storing such a program.
[0147] While the disclosure has been shown and described with reference to various embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the spirit and scope of the disclosure as defined by the appended claims and their equivalents.

Claims

1. An electronic device comprising: a communication circuit; memory, comprising one or more storage media, storing instructions; and at least one processor communicatively coupled to the communication circuit and the memory, wherein the instructions, when executed by the at least one processor, individually or collectively, cause the electronic device to: transmit, via the communication circuit, capability information of the electronic device and a request for transmitting capability information of an external electronic device to the external electronic device, obtain, via the communication circuit, the capability information of the external electronic device from the external electronic device, based on the obtained capability information of the external electronic device, generate preview content to be output from the external electronic device, and transmit, via the communication circuit, the generated preview content to the external electronic device.
2. The electronic device of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, further cause the electronic device to: search for an external electronic device for transmission of media content from the electronic device, and identify whether the generated preview content exists for the media content.
3. The electronic device of claim 2, wherein the instructions, when executed by the at least one

processor individually or collectively, further cause the electronic device to, in case that the generated preview content exists for the media content, process the generated preview content based on the capability information of the external electronic device.
4. The electronic device of claim 1, wherein the capability information received from the external electronic device comprises information related to at least one of a codec, frames per second (FPS), resolution, chipset, storage capacity, and network communication, of the external electronic device.
5. The electronic device of claim 1, wherein the capability information received from the external electronic device comprises information related to quality of the preview content that the external electronic device is capable of receiving.
6. The electronic device of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, further cause the electronic device to, based on the obtained capability information of the external electronic device, adjust an amount of the preview content to be transmitted to the external electronic device.
7. The electronic device of claim 2, wherein the media content comprises video content and image content, and wherein the instructions, when executed by the at least one processor individually or collectively, further cause the electronic device to reduce an amount of the preview content for the video content to be transmitted to the external electronic device in case that remaining capacity of the external electronic device is equal to or less than a predetermined amount of capacity, or that reception performance of the external electronic device is not capable of maintaining a quality of the generated preview content.
8. The electronic device of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, further cause the electronic device to: in case that an amount of preview content to be transmitted is adjusted, identify whether the preview content to be transmitted is capable of maintaining a predetermined quality on the external electronic device, in case that the predetermined quality is able to be maintained, transmit the preview content to be transmitted to the external electronic device via the communication circuit, and in case that the predetermined quality is unable to be maintained, process the preview content to be transmitted in a format supportable by the external electronic device.
9. The electronic device of claim 2, wherein the external electronic device comprises a plurality of external electronic devices, and wherein the instructions, when executed by the at least one processor individually or collectively, further cause the electronic device to: receive capability information for each of a plurality of reception devices, identify the media content to be transmitted to the plurality of reception devices, and in case that the media content to be transmitted is the same, produce the preview content based on a reception device having a lowest specification among the plurality of reception devices.
10. The electronic device of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, further cause the electronic device to receive the capability information for each of a plurality of reception devices within a predetermined time, and not transmit the preview content to corresponding reception device in case that the capability information is received after the predetermined time.
11. A method performed by an electronic device, the method comprising: transmitting, via a communication circuit of the electronic device, capability information of the electronic device and a request for transmitting capability information of an external electronic device to the external electronic device; obtaining, via the communication circuit, the capability information of the external electronic device from the external electronic device; based on the obtained capability information of the external electronic device, generating preview content to be output from the external electronic device; and transmitting, via the communication circuit, the generated preview content to the external electronic device.
12. The method of claim 11, further comprising: identifying whether the generated preview content exists for a media content.

13. The method of claim 12, further comprising: in case that the generated preview content exists for the media content, processing the generated preview content based on the capability information of the external electronic device.
14. The method of claim 11, wherein the capability information received from the external electronic device comprises information related to at least one of a codec, frames per second (FPS), resolution, chipset, storage capacity, and network communication, of the external electronic device.
15. The method of claim 11, wherein the capability information received from the external electronic device comprises information related to quality of the preview content that the external electronic device is capable of receiving.
16. The method of claim 11, further comprising, based on the obtained capability information of the external electronic device, adjusting an amount of the preview content to be transmitted to the external electronic device.
17. The method of claim 12, wherein the media content comprises video content and image content, and wherein the method further comprises reducing an amount of the preview content for the video content to be transmitted to the external electronic device in case that remaining capacity of the external electronic device is equal to or less than a predetermined amount of capacity, or that reception performance of the external electronic device is not capable of maintaining a quality of the generated preview content.
18. The method of claim 11, further comprising: in case that an amount of preview content to be transmitted is adjusted, identifying whether the preview content to be transmitted is capable of maintaining a predetermined quality on the external electronic device, in case that the predetermined quality is able to be maintained, transmitting the preview content to be transmitted to the external electronic device via the communication circuit, and in case that the predetermined quality is unable to be maintained, processing the preview content to be transmitted in a format supportable by the external electronic device.
19. One or more non-transitory computer-readable storage media storing one or more computer programs including computer-executable instructions that, when executed by one or more processors of an electronic device individually or collectively, cause the electronic device to perform operations, the operations comprising: transmitting, via a communication circuit of the electronic device, capability information of the electronic device and a request for transmitting capability information of an external electronic device to the external electronic device; obtaining, via the communication circuit, the capability information of the external electronic device from the external electronic device; based on the obtained capability information of the external electronic device, generating preview content to be output from the external electronic device; and transmitting, via the communication circuit, the generated preview content to the external electronic device.
20. The one or more non-transitory computer-readable storage media of claim 19, the operations further comprising: identifying whether the generated preview content exists for a media content.
